

**Domain-general ability underlies complex object ensemble processing**

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### **Abstract**

When seeing groups of objects, various features can be extracted to form an ensemble representation, including low-level features such as orientation and higher-level features like facial expression. Past research proposed distinct abilities for ensemble coding of high-level vs. low-level visual features, but the only complex objects used were faces. Here, we examine evidence for a shared ability supporting ensemble representations for complex objects from different object categories. In two experiments, participants completed an ensemble mean judgment task for an array of four objects, including planes and birds (Experiment 1) or cars and birds (Experiment 2). We also measured and controlled for domain-specific recognition ability. Across the two experiments, performance on ensemble judgments with different objects were correlated, even after controlled for domain-specific recognition abilities. These findings provide the strongest evidence to date of a domain-general ability involved in complex object ensemble coding.

*Keywords:* ensemble perception, individual differences, object recognition, summary statistics

## Introduction

When dealing with the complexity of visual information in a scene, humans can extract summary statistics from groups of similar objects, such as the average or variance of the color, size, or identity of these ensembles (see Whitney & Leib, 2018 for a recent review). Such judgments are thought to be supported by ensemble coding mechanisms that can compensate for limitations in perception and cognition (e.g., Bulakowski et al., 2011; Alvarez & Oliva, 2009; Haberman & Whitney, 2011; Im & Chong, 2014). Ensemble perception can occur at various levels of visual analysis, from lower-level features like orientation or motion direction (Dakin & Watt, 1997; Watamaniuk et al., 1989) to higher-level features such as facial emotion (Haberman & Whitney, 2007) and facial identity (de Fockert & Wolfenstein, 2009).

Even though ensemble coding requires processing of some information from individual objects, it has been argued to operate somewhat independently of object recognition. For example, ensemble perception remains robust when stimulus presentation is so short that individual items cannot be perceived (e.g., Haberman & Whitney, 2009, 2011) and judgments on ensembles tend to be more accurate than those on single objects (Alvarez, 2011; Robitaille & Harris, 2011). Brain structures engaged in ensemble perception have been argued to dissociate from those representing individual objects (Cohen et al., 2016; Leib et al., 2012). Resolving whether different tasks rely on different systems is a notoriously difficult endeavor (e.g., Ross et al., 2014), but the study of individual differences can suggest evidence for a common underlying mechanism through correlated performance. Indeed, the abilities to make judgments about ensembles and single objects have sometimes been found to correlate (Haberman et al., 2015; Sweeny et al., 2015) with some arguing that they are not independent (Neumann et al., 2018).

Despite growing research on ensemble perception (an informal citation search on Google Scholar showed a tripling of yearly publications using the phrase “ensemble perception” between 2001 and 2019), it is still unclear what mechanisms support these judgments. In an influential study, Haberman et al. (2015) used an individual differences approach to relate performance in tasks where participants reported the average of a feature in an array of four objects or the value of that same feature for a single cued object. Performance for ensemble judgments was strongly correlated between low-level features (e.g., color of dots and orientation of gabors) and between high-level features (face emotion and identity), but little to no correlation was found across high-level and low-level features (e.g., facial identity and orientation of gabors). These results indicated at least two separate ensemble judgment abilities, for low- and high-level features. Other work, although relatively underpowered, suggested different ensemble coding mechanisms for low-level features (i.e., orientation) and mid-level features (i.e., size, Yörük & Boduroglu, 2020). These results suggest that ensemble judgments are not supported by one single domain-general mechanism, although how many different mechanisms are involved remains uncertain.

Here, we examine the relationship between ensemble judgments for different complex object categories. Carving the space of all possible combinations of summary statistics (e.g., average, variance, numerosity), features (e.g., size, color, identity, expression) and object type (e.g., simple shapes, complex objects in different domains) based on pairwise correlations may be impractical. However, principles for carving this space can be extracted both from interesting dissociations (e.g., some work argues that judgments of average and variability for the same stimuli are not correlated—Yang et al., 2018; although see Cha et al., in press), and from interesting associations. The latter is what we seek here.

Little is known about the relationship between ensemble judgments for different object categories. Haberman et al. (2015) suggested commonality between different so-called “high-

level” tasks, but the only two high-level features they tested, identity and emotional expression, both used faces. We stay away from faces here, because individual face recognition ability is atypical in its relationship with other object categories (Sunday et al., 2019; Ćepulić et al., 2018). But in measuring ensemble perception for several non-face categories, we indirectly provide evidence that ensemble coding of complex objects extends to categories for which we may not have prior experience processing as a group (see also in Cha et al., 2021; Sunday et al., in press). This makes it clear that any suggestion that ensemble coding with faces benefit from mechanisms specifically related to social processing would require evidence that is currently lacking. In the current work, we were inspired by recent evidence for domain-general object recognition abilities (Richler et al., 2017; 2019; Sunday et al., in press). In some of these studies, structural equation modeling revealed that more than 85% of the variance in performance for different tasks, with a range of novel and familiar objects, is accounted for by a higher-level factor. Importantly, this domain-general ability is found to be largely independent of general intelligence (Richler et al., 2019), and as such can offer additional predictive power in a range of tasks where visual skills are important. For instance, it contributes significant variance in learning performance in learning to detect tumors in chest radiographs, after controlling for general intelligence (Sunday et al., 2018). However, to the extent that ensemble coding differs from object recognition, we need to investigate whether, just as for object recognition, we can find evidence for domain-general ensemble coding for categories that are visually distinct.

In two experiments, participants judged the mean identity for arrays of birds and planes (Experiment 1), or birds and cars (Experiment 2). Because variance in ensemble coding may be shared with object recognition (Haberman et al., 2015; Sweeny et al., 2015; but see Leib et al., 2012), we asked if ensemble coding for different categories would be related even after controlling for object recognition performance using the Vanderbilt Expertise Test (VET, McGugin et al., 2012). In addition, controlling for VET performance for each category should also reduce the contribution of domain general-general object recognition ability (Richler et al., 2019) and of trait and state influences common across performance measures (e.g., motivation, intelligence, and conscientiousness). Our results generally support a domain-general mechanism for ensemble coding of complex objects.

## Experiment 1

### *Participants*

Seventy-two participants with US IP addresses (36 males and 36 females; mean age = 39.01,  $SD = 9.44$ ) were recruited online via Amazon Mechanical Turk. These participants also had to meet our criteria of having more than 100 approved HITs with an approval rate of more than 95%. We based a power analysis on the effect size for the correlation among two different face tasks in Haberman et al. (2015), Experiment 7,  $r = .42$ . Forty-two participants would be needed for 80% power at an alpha of .05 (note however that we use Bayesian statistics, which are less dependent on power analysis). We collected more data to allow rejecting participants performing at chance and possibly inflating correlations. After dropping participants at chance on one task or who failed an attention check (described below), 53 participants remained (26 males and 27 females; mean age = 39.42,  $SD = 10.13$ ). The study was approved by the Vanderbilt Institutional Review Board and informed consent was obtained at the start of study. Participants received monetary compensation.

### *Stimuli, Design, and Procedure*

All participants completed two tests for each category (birds and planes): an ensemble

mean judgment task (EMJ) and a Vanderbilt Expertise Test (VET). To avoid confounds due to task order, all participants completed the tests in the same order<sup>1</sup>: Bird EMJ, Plane EMJ, Bird VET, Plane VET.

Ensemble mean judgment tasks (EMJ). The stimuli for each category were morphs created by linearly interpolating pairs out of three distinct exemplar models of that category using Creaseed Morph Age 4.2.4 (Figure 1). Six morphs (and none of the original images) were used in the tests. All objects were grayscale on a solid gray background.



Figure 1. The morph space for birds (Experiments 1 and 2), planes (Experiment 1), and cars (Experiment 2). Six images of passerine bird morphs were generated from photographs of Eastern Phoebe, Tree Swallow, and Yellow Throat (i.e., 25% Eastern Phoebe – 75% Tree Swallow, 75% Eastern Phoebe – 25% Tree Swallow, 25% Eastern Phoebe – 75% Yellow Throat, 75% Eastern Phoebe – 25% Yellow Throat, 25% Tree Swallow – 75% Yellow Throat, and 75% Tree Swallow – 25% Yellow Throat). Similarly, six plane morphs were generated from aircraft models from Gulfstream, Bell, and Bombardier Challenger. Six car morphs were generated from images of a BMW M6, a Chevrolet Spark, and a Jaguar F-Type.

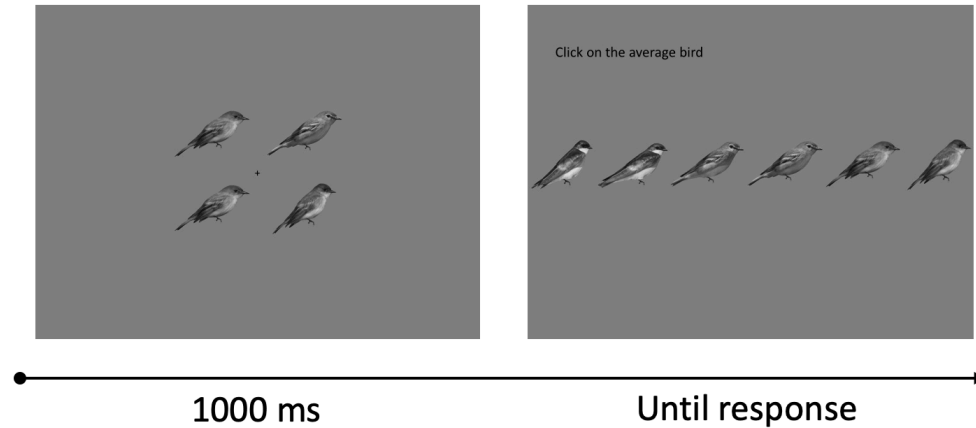
On each trial, a display with four items was shown for 1 second (Figure 2). Participants were asked to maintain fixation while studying the items. Afterwards, a response display with an array of the six morphs was shown – the morphs were ordered but started on a randomly selected item on each trial. Participants reported which of the six morphs was closest to the average identity of the four items by clicking. Five practice trials with feedback using the same six morphs were given (not analyzed), and no feedback afterwards. All participants completed 90 trials for each category.

Each trial included two identical morphs together with the two immediate flanking morphs from the response set. Each trial was unique in its combination of images and locations. Because a morph continuum may be heterogenous with mathematical contributions from original images not translating exactly into perceptual similarity, no item necessarily corresponded to the exact average identity (unless the four studied items are identical, which was never the case).

<sup>1</sup> This approach, standard in individual differences studies, hinges on the fact that we are not interested in the relative difficulty of the task. However, the approach does not control for possible interactions between task order and ability levels – a much larger dataset would be required to detect such effects.

Accordingly, we scored performance by calculating distance from the most correct choice, which was the repeated item on each trial. For every trial, we calculated the absolute value of the difference between the most correct choice and the response (e.g., if the participant responded 3 when the most correct choice was 4, their performance for that trial was scored as  $|3 - 4| = 1$ ). We multiplied distance by  $-1$  so that larger values reflected better performance (chance = 1.5).

### Ensemble Mean Judgment



### Vanderbilt Expertise Test

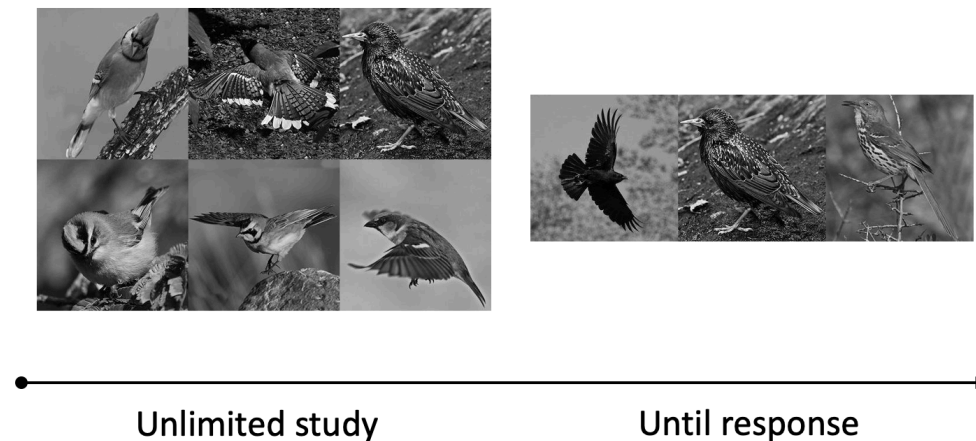


Figure 2. Example trials in the Mean Judgment Test (top panel) and the Vanderbilt Expertise Test (bottom panel). The trial procedures were the same for all object categories.

Vanderbilt Expertise Tests (VET). All participants completed the Bird VET followed by the Plane VET (McGugin et al., 2012). In each test, participants studied an array of six targets for as long as they needed, and completed a total of 48 3-alternative-forced-choice (3-AFC) trials, choosing which of 3 items was one of the 6 studied targets. The first 12 trials showed images identical to the target images and provided feedback. Following an opportunity to study the targets again, 36 trials with new images were included, without feedback. Three easy catch trials with very different foils (wading birds or sailboats) were included to ensure attention to

instructions – two participants made a mistake on more than one of those and were excluded. Average accuracy over the 48 experimental trials was calculated to index performance on these tests (chance = .33; see Sunday et al., 2018, for how all trials point to a unidimensional ability).

### *Results*

To avoid inflated correlations due to poor motivation, we excluded 17 participants who did not perform above chance on all tasks, leaving 53 participants. However, analyses conducted with the entire sample produced qualitatively similar results (aside from the Plane VET- Plane EMJ correlation for which we found support in that case,  $r = .33$ ,  $BF_{10} = 6.3$ ). Our measurements are reliable (Table 1), but with internal consistency around .8, this is also the effective ceiling for all expected correlations across tasks. Table 2 presents zero-order correlations between tasks (see Supplemental Materials for scatterplots for these correlations; Spearman rank-order correlations and skipped correlations are presented in Supplemental Materials and indicate no influence of bivariate outliers). We report Bayes factor ( $BF_{10}$ ) in favor of the presence of a correlation (relative to no correlation). In line with Jeffreys (1939), we consider a BF between 1 and 3 inconclusive evidence, a BF between 3 and 10 is moderate evidence, and a BF greater than 10, strong evidence. BFs below .33 are therefore considered evidence in favor of the null.

Table 1. Summary statistics for each test in each Experiment.

**Experiment 1**

|                      | Mean (SD)  | $\alpha$ | Skewness | Kurtosis | Range       |
|----------------------|------------|----------|----------|----------|-------------|
| Bird EMJ (distance)  | 0.95 (.19) | .81      | 0.36     | -0.25    | [0.66-1.42] |
| Plane EMJ (distance) | 1.10 (.24) | .86      | -0.21    | -0.97    | [0.59-1.49] |
| Bird VET (accuracy)  | .64 (.15)  | .85      | -0.05    | -0.66    | [.35-.92]   |
| Plane VET (accuracy) | .65 (.14)  | .80      | 0.13     | -0.50    | [.39-.98]   |

**Experiment 2**

|                     | Mean (SD)  | $\alpha$ | Skewness | Kurtosis | Range       |
|---------------------|------------|----------|----------|----------|-------------|
| Bird EMJ (distance) | 0.98 (.21) | .79      | 0.63     | -0.39    | [0.63-1.47] |
| Car EMJ (distance)  | 1.05 (.23) | .82      | 0.38     | -0.88    | [0.59-1.46] |
| Bird VET (accuracy) | .63 (.13)  | .78      | -0.25    | -0.68    | [.35-.88]   |
| Car VET (accuracy)  | .54 (.14)  | .82      | 0.58     | -0.44    | [.34-.90]   |

Table 2. Zero-order correlations for each Experiment, with  $BF_{10}$  in favor of a correlation over no correlation.**Experiment 1** Correlation ( $BF_{10}$ )

|           | Bird EMJ    | Plane EMJ | Bird VET  |
|-----------|-------------|-----------|-----------|
| Plane EMJ | .63 (34710) |           |           |
| Bird VET  | .48 (114)   | .48 (114) |           |
| Plane VET | .35 (4.0)   | .23 (.64) | .36 (4.6) |

**Experiment 2** Correlation ( $BF_{10}$ )

|          | Bird EMJ     | Car EMJ    | Bird VET    |
|----------|--------------|------------|-------------|
| Car EMJ  | .69 (1.67e7) |            |             |
| Bird VET | .41 (31.3)   | .33 (4.48) |             |
| Car VET  | .40 (20.8)   | .38 (12.7) | .47 (211.5) |

All tasks were correlated with at least moderate evidence, except for the Plane EMJ and the Plane VET (where there is inconclusive evidence for a correlation). Most interestingly, the strongest correlation was between the two EMJ tasks ( $r = .63$ ,  $BF_{10} = 34710$ ). To provide a stronger test for the possibility of a domain-general ability common to EMJ tasks, we regressed variability for each VET from the corresponding EMJ task, to obtain residuals that have a correlation of 0 with the VET from the same category. Accordingly, the high correlation between residuals for the Bird EMJ and the Plane EMJ tasks ( $r = .41$ ,  $BF_{10} = 17.47$ ) cannot be attributed to variance shared across VETs (Figure 3A). We also considered whether study time of the targets (under the participants' control) during the VET could have complicated the results (see Supplemental Materials). Study time is sometimes, but not always, related to VET scores. Controlling for study time in addition to the VETs did not reduce the correlation between the Bird EMJ and the Plane EMJ tasks ( $r = .44$ ,  $BF_{10} = 37.74$ ).



These results provide strong evidence of shared variance for ensemble tasks for two very different categories of objects, after controlling for domain-specific object recognition ability. One caveat is the inconclusive evidence of a relation between the Plane EMJ and the Plane VET. The EMJ tasks used a smaller set of items than the VET so it is possible that the Plane EMJ, by chance, did not have sufficient content validity, or that its content coverage was not overlapping with that of the Plane VET<sup>2</sup>. This limits our ability to control for individual differences in plane recognition ability. We conducted Experiment 2, replacing planes with cars, hoping car tests would behave more like bird tests.

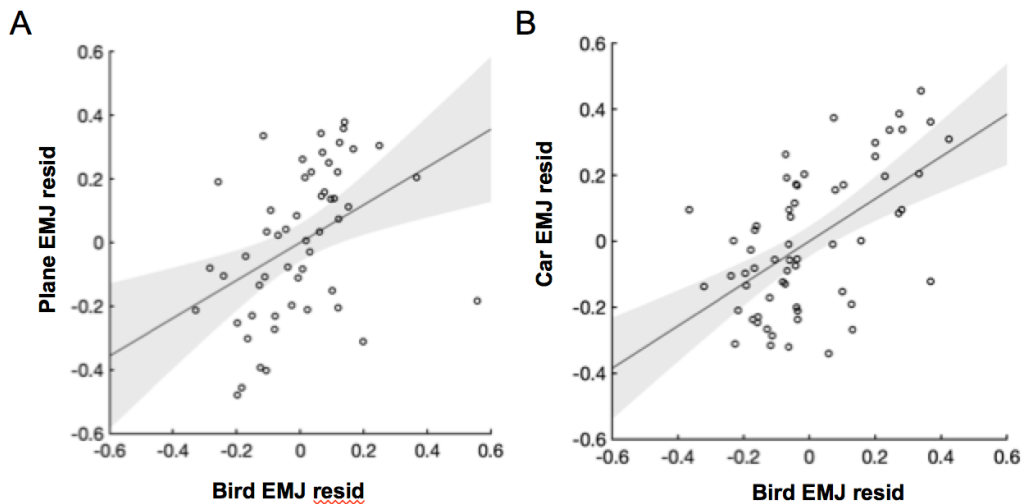


Figure 3. Partial correlation between performance in the two EMJ tasks, controlling for the VET scores, in Experiment 1 (A) and 2 (B).

## Experiment 2

### *Participants*

Seventy-three participants with US IP addresses (41 males and 32 females; mean age = 38.51,  $SD = 11.73$ ) were recruited online via Amazon Mechanical Turk. Same as in Experiment 1, these participants all had been approved for more than 100 approved HITs and for more than 95% of the HITs they had submitted. None of them had completed Experiment 1. Sixty-one participants (32 males and 29 females; mean age = 38.52,  $SD = 11.16$ ) performed above chance on all tasks and were included in further analyses. The study was approved by the Vanderbilt Institutional Review Board and informed consent was obtained at the start of study. Participants received monetary compensation for their participation.

### *Stimuli, Design, and Procedure*

The design was identical to Experiment 1 except that the two plane tests were replaced by the Car VET and a Car EMJ task (Figures 1 and 2). In addition, the EMJ tasks included 70 trials

<sup>2</sup> The choice of specific exemplars for the EMJ tasks is guided by morphing constraints and we note that finding plane images that would produce acceptable morphs was more difficult than for the other categories.

instead of 90 trials, as item analyses suggested this would be sufficient for reliable estimates. In the Car VET, three catch trials used antique cars and jeeps for attention check (no subject missed more than one of those).

### Results

All tasks were positively correlated with at least moderate evidence and, in most cases, strong evidence (see Table 2 and Supplemental Materials for relevant correlations and scatterplots). As expected, the Car EMJ task was correlated with the Car VET. Car recognition tasks have often (but not always, see Sunday, Lee, & Gauthier, 2018) shown low correlations with other object recognition tasks (McGugin et al., 2012; Ćepulić et al., 2018), but here we found the VETs for cars and birds to show a robust correlation ( $r = .47$ ,  $BF_{10} = 211.5$ ). As in Experiment 1, the strongest correlation was between the two EMJ tasks ( $r = .69$ ,  $BF_{10} = 1.67e7$ ). Again, we regressed variability from each VET from the corresponding EMJ task to obtain residuals that have a correlation of 0 with the VET from the same category. Accordingly, the high correlation between residuals for the Bird EMJ and the Car EMJ tasks ( $r = .57$ ,  $BF_{10} = 9631.9$ ) cannot be attributed to variance shared across VETs (Figure 3B). As in Experiment 1, controlling for VET study time in addition to VET scores did not substantially reduce the correlation between the two EMJ tasks ( $r = .56$ ,  $BF_{10} = 6241.9$ ).

### General Discussion

Haberman et al. (2015) studied individual differences to carve out the space of mechanisms that support ensemble coding judgments. Their results supported many conclusions, several of which we considerably expand on here. First, performance on single item judgments was related to performance on EMJ tasks, for both gabor orientation judgments and for face identity, although that correlation was stronger for faces than for gabors (.76 vs. .43). Here, we extend these within-domain correlations across object recognition and EMJ tasks to new categories. When ensemble or single item judgments following the presentation of identical arrays of items (as in Haberman et al., 2015), a correlation could stem from variance driven by the specific stimuli. At least one study found no correlation between ensemble judgments for faces and separate tests of face recognition (Leib et al., 2012), so it is particularly important that in the VETs, objects appeared in various viewpoints and on cluttered backgrounds, using completely different exemplars from those used to create morphs for the EMJ tasks. Our results therefore ensure that similarity in task format or overlap in the specific objects used is not driving the correlations between ensemble judgments and object recognition.

Moreover, because our experiments used more than one category of complex object, we can show that correlations across tasks *and* across categories (.48, .35, .33, .40; Fisher-transformed average: .32) were comparable in magnitude to those across tasks but *within* category (.48, .23, .41, .38; Fisher-transformed average: .33). This suggests that the source of the overlap in resources across EMJ and single item judgment is domain-general object recognition rather than category-specific. In other words, when the Bird EMJ correlates with the Car VET, it is not due to items, task format, nor category, all aspects that could have contributed to correlations in prior work. We propose that the source of this shared variance is *o*, or domain-general object recognition ability, as recently characterized using structural equation modeling of performance with several tasks and several object categories (Richler et al., 2019). Haberman's finding of essentially null correlations even within tasks but across faces and simple features

(gabor orientation or color), would further suggest that while *o* may apply across domains, it may not extend to decisions about simple features.

Second, Haberman et al. (2015) found that judgments of average identity and average emotion for faces were correlated, as were different ensemble judgments for simple features (e.g., gabor orientation and color). In contrast, they found very small and non-significant correlations across different ensemble tasks for faces and simple features. This led to the idea of at least two separate mechanisms for ensemble coding, for low- and high-level judgments. However, because their only complex category was faces, the evidence for convergence of an ensemble mechanism for high-level category was much weaker than for low-level features. In fact, other work has suggested that even for faces, judgments of average identity and average viewpoint can be dissociated (Sama et al., 2019). We demonstrate for the first time that EMJ tasks with different high-level categories are strongly correlated, and that this correlation is not fully explained by object recognition ability. Our results converge with others (Cha et al., in press) to demonstrate that ensemble judgments are possible for non-face complex objects – and that experience making such judgments for a category is not required (to the extent we assume most of our participants rarely need to average bird, car or plane identities). This raises the bar for thinking about ensemble coding for categories where people may have such experience. Finding that people can perceive ensembles of faces does not directly indicate that their skill is especially impacted by its relevance for social interactions.

There remains considerable work to do to understand the mechanisms that support different kinds of ensemble judgments and how they relate to other aspects of cognition. A clearer picture will require combining tests for the kinds of complex categories that we tested with the kinds of low-level tasks used by Haberman et al. (2015). Our results suggest that we do not need to invoke different ensemble coding mechanisms for different object categories, just like the work on *o* suggests it is not necessary to invoke different object recognition mechanisms for different object categories (Richler et al., 2019)<sup>3</sup>. We conclude that ensemble coding relies in part on a mechanism that is different from object recognition, but more extensive datasets and latent variable modeling could provide additional tests of this hypothesis (Richler et al., 2019; Wilhelm et al., 2010). Findings that ensemble judgments can be more precise than judgments on single items (Robitaille & Harris, 2011), or that they can be made under conditions where individual judgments are at chance (e.g., Haberman & Whitney, 2009, 2011) could mean that the same mechanism is involved but can operate more efficiently when people sample from several items. Stronger claims for distinct mechanisms will require explicit modeling of the processes involved (Nosofsky & Johansen, 2000). Ensemble coding of object identity, even if it proves to be distinct from object recognition, appears to depend on a domain-general ability, as was recently reported for object recognition.

### Context of the Research

This research was motivated by the current scientific interest in the nature of mechanisms underlying ensemble perception. Ensemble judgments have been measured for various tasks and a wide array of simple stimuli. However, few studies have used complex objects other than faces. We were specifically inspired by the Haberman et al. (2015) study, which suggested two broad abilities for mean judgments of low-level features (e.g., orientation and color) vs. high-level

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<sup>3</sup> There may be exceptions to this rule when subjects vary a great deal in experience or have very high levels of experience with a category (Sunday, Lee, & Gauthier, 2018).

features (e.g., facial expression and face identity), while the mean judgments of different low-level (or high-level) features were correlated with each other. However, the only two high-level tasks that were examined used the same category, faces. Here, we expand on these findings and assess evidence of a common ability underlying different mean identity judgments for complex objects from different categories. Our results provide strong evidence that a domain-general ability contributes to ensemble coding of complex objects. It also offers a clear demonstration that ensemble coding for faces is not a special ability that requires expertise or relies on social knowledge. In future work, we will continue to develop and compare different ensemble perception tasks for complex objects with good psychometric properties and explore the relative variance explained by properties of the task, the specific summary statistic that is judged, and domain-general vs. domain-specific skills.

Open Practices Statement:

None of the experiments were preregistered, the data are available online (<https://figshare.com/s/15b97a3443560bcde028>) and the materials are available upon request.

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